

Data 607 Project 4

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Approach

For this assignment, I would begin by choosing a labeled dataset of spam and ham emails, such as the Apache SpamAssassin dataset. Since the emails are already labeled, this makes it easier to train a model to recognize patterns in spam versus non-spam messages. First, I would load the emails into R and organize them into a dataset with two categories: spam and ham.

Next, I would clean the text data so it can be analyzed properly. This would include converting all words to lowercase, removing punctuation, numbers, extra spaces, and common words such as “the” or “and.” After cleaning the text, I would turn the emails into numbers using methods like a document-term matrix or TF-IDF so the computer can measure word importance.

Then, I would split the dataset into a training set and a testing set. The training set would be used to teach the model how spam and ham emails differ. The testing set would be used afterward to see how well the model can classify new emails it has never seen before. I would likely test models such as Naive Bayes or Logistic Regression because they are commonly used for text classification.

To measure success, I would use accuracy, precision, recall, and a confusion matrix. These results would show how often the model correctly identifies spam and how often it mistakes regular emails for spam.

Some data challenges I may encounter include messy email text, special characters, HTML formatting, and duplicate emails. Another challenge is that spam emails may use unusual spellings or symbols to avoid detection. Some words may also appear in both spam and ham emails, making classification harder. I may also face unbalanced data if there are many more spam emails than ham emails or vice versa. To handle these issues, I would carefully clean the data, remove duplicates, and test different models to improve performance.

Link: <https://spamassassin.apache.org/old/publiccorpus/>